Band-1 Andrews–Curtis: AK(3) Directed Bounded Non-Finding

Bounded search non-finding / verification only (NOT AC solved; AK(3) OPEN; GATE-04 LOCKED)

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Document class. Scientific computational increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-11_Band1_Andrews_Curtis_presentation_probe`; `2026-09-14_Band1_Andrews_Curtis_AK2` `2026-09-14_Band1_Carreras_AC_length14_verify`.
Lab. `/workspace/experiments/band1/ak3_bounded_nonfinding/`
Cutoff. 2026-09-14 (~7:02PM ET).

Honesty stamp (non-negotiable).
This memo is a **bounded search non-finding / verification** increment only. It **does not** claim the Andrews–Curtis conjecture. It **does not** trivialize or refute **AK(3)**. A miss inside finite bounds is **not** a counterexample. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED** (no invented meter J). No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. `clay_closed = false`.

Abstract

We deepen the prior shallow AK(3) BFS non-finding (visited 11 176) with a **multi-bound directed grid** and an **AC-stabilized length-priority** (best-first on total relator length) heuristic on

$$\mathrm{AK}(3) = \langle a, b \mid a^3 b^{-4},\ abab^{-1}a^{-1}b^{-1} \rangle$$

(total length 13). Same self-contained AC1–AC3 code first re-confirms the lit-known **AK(2)** control: BFS **TRIVIALIZED** in 17 moves (visited 40 141; verify **PASS**). All AK(3) cells report **NOT_FOUND_IN_BOUNDS**; deepest BFS exhausts a frontier of 739 047 distinct normalized presentations (wall ≈ 39.6,s) with best total length stuck at 13. **AK(3)** and the full Andrews–Curtis conjecture remain **OPEN**.

Keywords. Andrews–Curtis; Akbulut–Kirby AK(3); bounded BFS; length-priority search; Band-1; non-finding; non-claim.

1. Statement lock

Andrews–Curtis (rank 2). A balanced presentation $\langle x, y \mid r, s \rangle$ of the trivial group is AC-trivializable if a finite sequence of elementary AC moves reaches $\langle x, y \mid x, y \rangle$ (up to free reduction).

Finite lemma (this increment only). Within the stated state / length / depth / wall-clock bounds, directed search on AK(3) does not reach a trivial presentation. **Not** AC; **not** AK(3) trivialized; **not** a counterexample.

2. What changed vs prior Band-1 AC probes

Axis	2026-09-11 probe	2026-09-14 AK(2) replay	Carreras length-14	**This increment**
AK(2)	shallow miss	BFS 17 + Havas 14 PASS	retained	**control TRIVIALIZED + PASS**
AK(3) shallow	miss	NOT_FOUND visited 11 176	—	superseded
AK(3) deep grid	—	—	—	**BFS shallow/medium/deep + LP medium/deep**
MS(3)∼AK(3)	lit	—	**5/5 PASS**	retained (equivalence ≠ trivialization)

Axis	2026-09-11 probe	2026-09-14 AK(2) replay	Carreras length-14	**This increment**
Claim level	methods	verification replay	verification	**bounded non-finding; AK(3) OPEN**

3. Methods

1. Self-contained copy of AC move machinery (`neighbors`, `bfs_trivialize`, `ak(n)`, `verify_sequence`, `normalize_presentation`, free reduction) into `probe_ak3.py` (no live import dependence on the AK(2) lab at runtime).
2. **AK(2) control** under identical code: BFS with `max_total_len=15`, `max_rel_len=15`, `max_depth=20`, `max_states=800\,000` — expect `TRIVIALIZED` + end-to-end `verify_sequence` **PASS**.
3. **BFS grid** on AK(3):
   - shallow: `max_states=100k`, `max_rel_len=14`, `max_depth=16`, `max_total_len=14`, wall $\leq$ 120,s
   - medium: `max_states=800k`, `max_rel_len=16`, `max_depth=24`, `max_total_len=18`, wall $\leq$ 150,s
   - deep: `max_states=2\,000\,000`, `max_rel_len=18`, `max_depth=30`, `max_total_len=20`, wall $\leq$ 180,s
4. **Length-priority** heuristic (Dijkstra-like heap keyed by `present_len`) at medium and deep caps — prefer neighbors that decrease / keep small total relator length; still a finite non-finding if miss.
5. Record per cell: status, visited, seen, best_len, best presentation, wall_s, bounds, termination reason.
6. Write `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `README.md`, this memo.

Elementary AC moves (box).

Tag	Action
AC1	$r_i := r_i\, r_j^{\pm 1}$ for $i \neq j$
AC2	$r_i := r_i^{-1}$
AC3	$r_i := g^{-1} r_i g$ with $\lvert g\rvert = 1$

Canonical trivial target. Both relators unit-length and hit $\{a^{\pm 1}, b^{\pm 1}\}$ exactly once each.

4. Results (box)

4.1 AK(2) control (sanity)

$$\mathrm{AK}(2) = \langle a, b \mid a^2 b^{-3},\ abab^{-1}a^{-1}b^{-1} \rangle$$

Metric	Value
Status	**TRIVIALIZED**
Moves	**17**
Visited / seen	40 141 / 58 699
Wall	≈ 2.27,s
`verify_sequence`	**PASS**

Same rediscovery envelope as the 2026-09-14 AK(2) replay lab (methods continuity; not a new math claim).

4.2 AK(3) BFS grid

$$\mathrm{AK}(3) = \langle a, b \mid a^3 b^{-4},\ abab^{-1}a^{-1}b^{-1} \rangle \quad \text{(Havas start } (aaaBBBB,\ abaBAB)\text{; total length 13)}$$

Cell	max_states	max_rel_len	max_depth	max_total_len	Status	Visited	Seen	Best len	Wall (s)	Reason
shallow	100 000	14	16	14	**NOT_FOUND**	42	42	13	0.002	exhausted_frontier
medium	800 000	16	24	18	**NOT_FOUND**	28 717	28 717	13	1.50	exhausted_frontier
deep	2 000 000	18	30	20	**NOT_FOUND**	739 047	739 047	13	39.56	exhausted_frontier

Shallow's tiny frontier is expected: from length-13 start, `max_total_len=14` / `max_rel_len=14` admits almost no AC1 growth (consistent with Havas-style length barriers observed for AK(2) at ≤ 12).

4.3 Length-priority (best-first on `present_len`)

Cell	Caps (as medium/deep)	Status	Visited	Seen	Best len	Wall (s)	Reason
lp_medium	medium	**NOT_FOUND**	23 685	23 685	13	1.17	exhausted_frontier
lp_deep	deep	**NOT_FOUND**	313 388	313 388	13	16.27	exhausted_frontier

No cell reached a shorter-than-13 presentation under these caps; best stayed at the start length. Total probe wall ≈ 61.3,s.

5. What remains OPEN

1. **Ordinary AC-trivialization of AK(3)** — OPEN.

2. Full **Andrews–Curtis** conjecture — OPEN.

3. Collapse of Carreras' two **MS(2)** open classes to AK(3) — OPEN (equivalence certs retained; $\neq$ trivialization).

4. Whether a trivialization exists outside these length / depth / state envelopes — out of scope; non-finding inside bounds is silent on existence.

6. Honesty / non-claims

Claim	Status in this memo
Andrews–Curtis conjecture solved	**NO**
AK(3) trivialized	**NO** — OPEN
AK(3) disproved / counterexample	**NO** (non-finding $\neq$ counterexample)
Clay Millennium stamp	**NO**
Finished ToE / GATE-04 collection	**NO** — GATE-04 **LOCKED**
RH / P$\neq$NP / YM / Frankl / NS solved	**NO**
`clay_closed`	**false**

7. Artifacts and reproducibility

Path	Role
`experiments/band1/ak3_bounded_nonfinding/probe_ak3.py`	Self-contained AC + BFS grid + length-priority
`.../results.json`	Machine-readable cell ledger
`.../literature_ledger.json`	Citations + OPEN entries
`.../SUMMARY.txt`	One-page human summary
`.../README.md`	Lab brief
Shelf PDF twin	`/workspace/DOWNLOAD_PDFs/2026-09-14_Band1_Andrews_Cu`

Reproduce. `python3 probe_ak3.py` on box; expect AK(2) control TRIVIALIZED+PASS and all AK(3) cells `NOT_FOUND_IN_BOUNDS` under the stated caps (wall clocks are soft caps; this run exhausted frontiers before hitting them).

8. Next optional (honest)

1. Ingest/build lit H_{508} then Miyamoto $\to H_{2036}$ (Hadamard track), **or** Track B privacy / Object 1 meter await (GATE-04).

2. Do **not** stamp AK(3) or full AC closed from deeper non-findings alone.

3. Optional future: import any public AK(3) *trivialization* certificate if one appears (verification only) — none claimed here.

References (selected)

1. J.J. Andrews & M.L. Curtis, Free groups and handlebodies, *Proc. Amer. Math. Soc.* **16** (1965) 192–195.

2. S. Akbulut & R. Kirby, A potential smooth counterexample…, *Topology* **24** (1985) 375–390.

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4. G. Havas & C. Ramsay, Breadth-first search and the Andrews–Curtis conjecture, *Internat. J. Algebra Comput.* **13**(1) (2003) 61–68.

5. J. Carreras, Machine-checkable Andrews–Curtis certificates…, arXiv:2607.23611 (length-14 MS; MS(3)∼AK(3)).

6. A. Shehper et al., What makes math problems hard for reinforcement learning…, arXiv:2408.15332.

End stamp. Bounded search non-finding / verification · AK(2) control PASS · AK(3) deep BFS visited 739 047 NOT_FOUND · best_len = 13 · AC **not** solved · AK(3) **OPEN** · Clay **not** stamped · GATE-04 **LOCKED** · `clay_closed=false` · 2026-09-14.